

Amjad S Althabteh

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C++ PROJECTS (github : AmjadAlthabteh)

3D Space Simulation Engine

[demo](#)

OrbitalCollision

- Built a modular C++17/OpenGL orbital collision prediction engine that modeled 50+ orbital bodies using Newtonian gravity, vector mathematics, and RK4 numerical integration across more than 1M trajectory calculations.
- Reduced simulation runtime by ~40% using spatial partitioning and multithreaded trajectory propagation, with SIMD-friendly, -O3-optimized math.
- Leveraged OpenGL, SFML, and ImGui to implement real-time rendering, visualization, and debugging tools for the simulation.

Glint 3D Rendering Engine | C++17/OpenGL

[demo](#)

Glint-Rendering

- Designed and implemented a custom C++ memory pool allocator using placement new, aligned storage, and free-list reuse, reducing repeated heap allocations and allocation overhead by approximately 60%.
- Optimized rendering performance using frustum culling and GPU instanced rendering, cutting draw calls by ~40%.
- Reduced cold-start latency by storing frequently accessed asset data in cache-friendly contiguous containers.

HFT Order Book | Low-Latency Performance System

HFT-lowlatency

- Built a C++20 low-latency order book using STL containers, move semantics, smart pointers, memory alignment, price-time priority, and unit-tested order matching for reliable trade execution.
- Reduced tail latency via 64-byte cache-aligned structures and false-sharing prevention, validated with p50/p99/p99.9 benchmarks under load.
- Benchmarked GCC vs. Clang across optimization levels and combined the best-performing flags into a custom build, beating both compilers' defaults on p99 latency.

IronPath Robotic Control Systems

robotic-ironpath

- Built a multithreaded C++ robotic control system using std::chrono, atomics, std::threads and CSV telemetry, sustaining a 100 Hz control loop with 142 μ s average latency and 540 μ s p99 latency.
- Developed a custom memory leak detector by overloading operator new and operator delete, improving memory diagnostics and application reliability.

EXPERIENCE

Graphics Engineer | Detroit, MI

05/25 – 02/26

Immersalab Graphics

- Built internal C++/JSON tooling to automate GLSL shader compilation and asset-processing pipelines, reducing manual build steps and improving iteration speed for the rendering team.
- Reduced GPU idle time by approximately 30% through RenderDoc profiling and pipeline optimizations, improving frame time, submission latency, throughput, and GPU utilization.
- Diagnosed and resolved shader compilation bottlenecks by profiling draw call submission order, cutting redundant state changes and improving frame consistency under load.

UE Game Programmer C++ (Contract) | Yorktown, VA

04/26 – 07/26

Grand Forge Games

- Built gameplay abilities in C++ using Unreal Engine's Gameplay Ability System (GAS) with Blueprint integration for game mechanics.
- Debugged and optimized animation blend graphs to eliminate stutter during transition blends, improving visual fidelity during fast-paced gameplay sequences.
- Improved animation performance using layered blending and reusable C++ gameplay systems, cutting per-ability implementation time for the design team.

Wayne State University | Detroit, MI

02/25 – 12/25

Undergraduate Researcher – Spacecraft Dynamics & Collision Analysis

- Evaluated GCC vs. Clang across all optimization levels, identifying GCC as the faster choice for simulation workloads while maintaining comparable binary size.
- Improved collision-analysis performance by approximately 30% using spatial partitioning, enabling faster large-scale simulations.
- Researched orbital trajectories and spacecraft collision risks across 10,000+ space objects using mathematical prediction models.

SKILLS

Languages: C++ (11–23), Rust, C, Python, Java, R, q/KDB+, Verilog, Scripting (Bash/PowerShell)

Graphics & Engines: OpenGL, GLSL, Unreal Engine, Unity, Godot, Vulkan

Systems & Hardware: Performance, multithreading, concurrency, pools, cache optimization, profiling, FPGA, Linux, GDB

EDUCATION

Wayne State University

B.S. in Computer Science | Minor in Mathematics | Top 6% in Class | GPA: 3.78

Detroit, MI

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